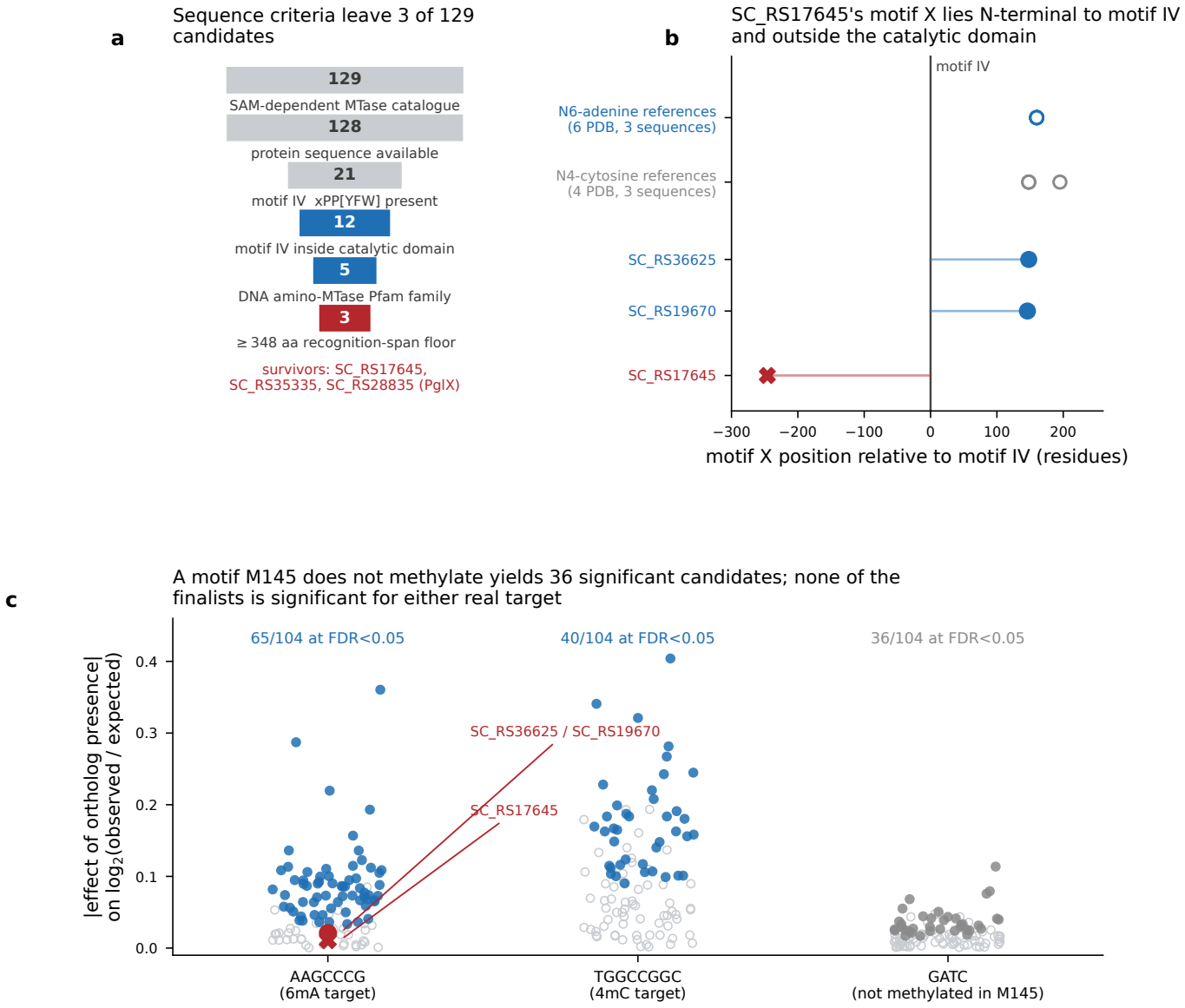
a
Sequence criteria leave 3 of 129 candidates
129
SAM-dependent MTase catalogue
128
protein sequence available
21
motif IV xPP[YFW] present
12
motif IV inside catalytic domain
5
DNA amino-MTase Pfam family
3
≥ 348 aa recognition-span floor
survivors: SC_RS17645, SC_RS35335, SC_RS28835 (PglX)
b
SC_RS17645's motif X lies N-terminal to motif IV and outside the catalytic domain
motif IV
N6-adenine references (6 PDB, 3 sequences)
N4-cytosine references (4 PDB, 3 sequences)
SC_RS36625
SC_RS19670
SC_RS17645
−300
−200
−100
0
100
200
motif X position relative to motif IV (residues)
c
A motif M145 does not methylate yields 36 significant candidates; none of the finalists is significant for either real target
65/104 at FDR<0.05
40/104 at FDR<0.05
36/104 at FDR<0.05
SC_RS36625 / SC_RS19670
SC_RS17645
|effect of ortholog presence| on log2(observed / expected)
0.0
0.1
0.2
0.3
0.4
AAGCCCG (6mA target)
TGGCCGGC (4mC target)
GATC (not methylated in M145)